

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0407

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.
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Code of Conduct:

*I **P. DEEKSHANA / 192425104** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



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CO1 – AT2 – ERROR ANALYSIS TASK

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COURSE CODE: MLA0407

COURSE NAME: DEEP LEARNING

Introduction:-

Mean Squared Error (MSE) is a performance metric used to measure the average squared difference between the actual and predicted values of a model. A lower MSE indicates better prediction accuracy. In this problem:

- Training MSE = 0.85
- Testing MSE = 5.20

The model performs well on the training data but poorly on the testing data. This large difference indicates that the model is unable to generalize well to unseen data.

1. Error Identification

The significant gap between the training and testing MSE indicates overfitting.

Overfitting occurs when the neural network learns the training data too closely, including its noise and unnecessary details, instead of learning the general patterns. As a result, it achieves very low training error but produces high error on new data.

2. Error Analysis

The possible reasons for the high testing error are:

a) Overfitting

The model memorizes the training data instead of learning meaningful relationships. This results in excellent training performance but poor testing performance.

b) Insufficient Training Data

If the training dataset is small or not diverse enough, the model cannot learn all possible patterns, reducing its ability to predict unseen data accurately.

c) Poor Data Quality

Missing values, noisy data, duplicate records, and outliers can negatively affect the learning process and increase prediction errors.

d) High Model Complexity

Using too many hidden layers or neurons makes the model unnecessarily complex, increasing the risk of overfitting.

e) Excessive Training

Training the model for too many epochs may cause it to memorize the training data rather than learning general features.

3. Corrective Measures

The following methods can improve the model's performance:

- Apply L1/L2 Regularization to reduce overfitting by controlling large weight values.
- Use Dropout to randomly deactivate neurons during training, improving generalization.
- Implement Early Stopping to stop training when the validation error begins to increase.
- Increase the size and diversity of the training dataset.
- Perform Data Preprocessing by handling missing values, removing outliers, and normalizing the data.
- Reduce the number of hidden layers or neurons if the model is too complex.
- Use Cross-Validation to evaluate the model on different data splits and improve reliability.
- Tune hyperparameters such as the learning rate, batch size, and number of epochs to achieve better performance.

4. Interpretation of Results

The low training MSE (0.85) indicates that the model has learned the training data effectively. However, the high testing MSE (5.20) shows that it fails to perform well on unseen data, which is a clear sign of overfitting.

By applying the suggested corrective measures, the testing error can be reduced, allowing the model to generalize better and make more accurate predictions for new student data.

Conclusion:-

The neural network suffers from overfitting, as shown by the large difference between the training and testing MSE values. The main causes include excessive model complexity, insufficient or poor-quality training data, and overtraining. Techniques such as Regularization, Dropout, Early Stopping, Data Preprocessing, Cross-Validation, and Hyperparameter Tuning can significantly reduce the testing error and improve the model's overall accuracy and reliability. This enables the neural network to make accurate predictions on both training and unseen data.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____